

Difference-in-Differences

Staggered Adoption: One Question, Three Assumptions

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Where we are

The previous chapter had **one treatment date**: a treated group, a control group, before and after. The event study put one coefficient per period; parallel trends was one assumption.

Most policies occur in stages. They **roll out**: some units adopt early, some late, some never. Minimum wages by state in the US, broadband by municipalities, emission-control areas by sea, organic certification by farm.

This chapter

Under staggered adoption the two-way fixed-effects regression mixes clean comparisons with comparisons to already-treated units, and its coefficient can be biased by treatment dynamics alone. This chapter shows why, and replaces the regression by an explicit construction: group-time effects identified under a stated parallel-trends assumption, then aggregated with transparent weights.

Reference for the whole chapter: Baker, Callaway, Cunningham, Goodman-Bacon & Sant'Anna (2026), "Difference-in-Differences Designs: A Practitioner's Guide".

Roadmap

- ➊ **The setting** — cohorts, event time, never- and not-yet-treated units.
- ➋ **What TWFE computes** — a weighted average of 2×2 comparisons, some forbidden.
- ➌ **The dynamic TWFE is worse** — contaminated leads and lags.
- ➍ **One question, three assumptions** — $ATT(g, t)$, the three parallel-trends assumptions, and the estimators as their sample analogues.
- ➎ **Aggregation, diagnostics, workflow.**

Section 1

Setting

Units adopt at different times

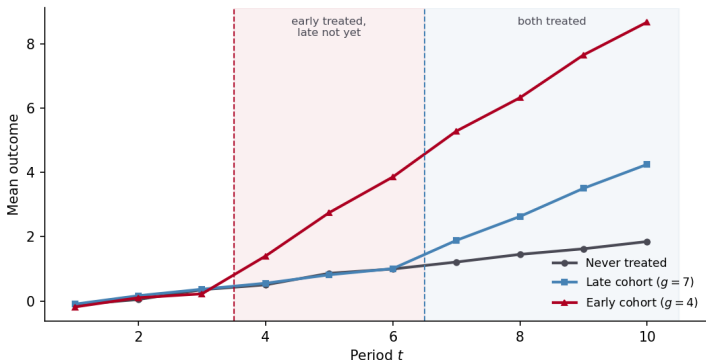
Unit i is first treated at date g_i — its **cohort**. Units never treated in the sample have $g_i = \infty$. **Event time** is $k = t - g_i$.

- Example — Emission Control Areas: Baltic (2006), North Sea (2007), North America (2012), Caribbean (2014).
- At any date t , a unit is **treated** ($t \geq g_i$), **not yet treated** ($t < g_i < \infty$), or **never treated**. The last two are the candidate comparison groups.
- Potential outcomes now carry the date: $Y_{it}(g)$ is the outcome if first treated at g ; $Y_{it}(\infty)$ if never. No-anticipation: $Y_{it}(g) = Y_{it}(\infty)$ for $t < g$.
- The natural regression is the **static TWFE** of the previous chapter with one switch,

$$y_{it} = f_i + f_t + \beta d_{it} + u_{it}, \quad d_{it} = \mathbf{1}\{t \geq g_i\},$$

and its dynamic version with leads and lags of k . **This is where it breaks.**

A simulated roll-out



Three cohorts, effects that **grow** with time since treatment and differ across cohorts. In periods 4–6 the late cohort is a clean comparison for the early one. In periods 7–10 TWFE still compares the late cohort to the early one, which is already treated and still responding: its change is not a counterfactual — a **forbidden comparison**.

Section 2

What TWFE computes

Two periods, three groups: the algebra

Two periods, three groups: already treated ($g = 1$), treated in period 2 ($g = 2$), never ($g = \infty$). In period 2 the switch is 1 only for $g = 2$: TWFE is a weighted average of **two** DiDs,

$$\beta_{\text{twfe}} = (1 - w) [\mathbb{E}(Y_2 \mid g=2) - \mathbb{E}(Y_2 \mid g=\infty)] + w [\mathbb{E}(Y_2 \mid g=2) - \mathbb{E}(Y_2 \mid g=1)].$$

The first bracket is $\text{ATT}(2, 2)$ under parallel trends. The second — comparison group **already treated** — equals $\text{ATT}(2, 2) - [\text{ATT}(1, 2) - \text{ATT}(1, 1)]$ (add and subtract, no-anticipation, parallel trends).

The estimand (Baker et al. 2026, eq. 5.17)

$$\beta_{\text{twfe}} = \text{ATT}(2, 2) + w \text{ATT}(1, 1) - w \text{ATT}(1, 2).$$

$\text{ATT}(1, 2)$ enters with a **negative** weight unless effects do not change over time. Dynamics alone — no failure of parallel trends — bias TWFE.

Goodman-Bacon (2021): the general decomposition

With many periods and cohorts, TWFE is a weighted average of **all** the 2×2 DiDs the data allow — early vs never, late vs never, late vs early (clean, early not yet treated), and **early vs late-as-control** (forbidden: the “control” is already treated and moving):

$$\hat{\beta}_{\text{TWFE}} = \sum_k w_k \widehat{\text{DiD}}_k, \quad \sum_k w_k = 1.$$

- The weights are **positive** and mechanical: larger for larger groups and for cohorts treated near the **middle** of the panel (largest variance of d_{it} after the fixed effects). They are unrelated to the policy question.
- The bias lives in the forbidden terms: each contains $-[\text{ATT}(g, t) - \text{ATT}(g, g)]$ for the already-treated cohort, as on the previous slide.
- de Chaisemartin & D’Haultfœuille (2020): on the *estimand*, some $\text{ATT}(g, t)$ get **negative weights** and β can take the wrong sign — the course’s recurring theme (compliers in IV, Rotemberg weights in shift-share).

Example: unilateral divorce and suicide

Unilateral divorce and women's suicide (Goodman-Bacon 2021)

US states adopted unilateral divorce at staggered dates (California 1969 ...; a few never). Re-analysing Stevenson & Wolfers (2006), Goodman-Bacon shows the **TWFE** estimate of the fall in women's suicide is **smaller** than the timing-robust one (−3.1 vs −7.0).

- The already-treated comparisons **attenuate** the effect toward zero — here the bias shrinks the estimate rather than flipping its sign (both are possible).
- The decomposition is a **diagnostic** to report: which comparisons carry the weight, and how much of β comes from the forbidden ones.

Section 3

Dynamic TWFE

The event study inherits the problem

The usual fix is to replace the switch by leads and lags, $\sum_{k \neq -1} \beta_k \mathbf{1}\{t - g_i = k\}$, one coefficient per event time.

Sun and Abraham (2021)

Under staggered timing, each $\hat{\beta}_k$ is a weighted average of $\text{ATT}(g, g + k)$ and of effects at **other** event times from **other** cohorts — with weights that can be negative. The lead coefficients ($k < 0$) are contaminated by post-treatment effects: a "pre-trend" can appear *because of* the treatment.

- The contamination comes from the same source: cohorts serve as each other's controls while they are themselves responding.
- It vanishes only if effects are homogeneous across cohorts — precisely what an event study is meant to explore.

Section 4

Three assumptions

The building block, again

The modern approach does not start from a regression. It starts from the parameter:

$$\text{ATT}(g, t) = \mathbb{E}[Y_{it}(g) - Y_{it}(\infty) \mid g_i = g], \quad t \geq g,$$

the effect at date t for the cohort first treated at g — a **group-specific event study**.

Forward engineering (Baker et al. 2026)

Every $\text{ATT}(g, t)$ is identified by a 2×2 : cohort g against a **comparison group**, period t against a **base period**. A staggered design is a collection of such 2×2 s, then an aggregation. All the tools of the previous chapter — pre-trends, placebos, balance, clustering — carry over block by block.

Two choices remain, and they are **not** software choices: *which comparison group* and *which base period*. Both are dictated by which parallel-trends assumption you are willing to make.

Three parallel-trends assumptions

	Untreated trends equal for...	Comparison	Base period
PT-never	cohort g and the never-treated , post-periods only	never-treated	$g - 1$ only
PT-not-yet	cohort g and any unit not yet treated at t , post-periods only	never- and not-yet-treated	$g - 1$ only
PT-all	every pair of cohorts in every period, pre and post	any not-yet-treated	any pre-period (or their average)

- Under PT-never or PT-not-yet, parallel trends is imposed **only after** treatment; the only justifiable base period is $g - 1$. Pre-trends are *evidence*, not part of the assumption.
- Under PT-all, parallel **pre-trends** are part of the assumption — which buys precision and makes pre-trends **testable** (the model is over-identified).

Estimators are sample analogues of an assumption

Estimator	Assumption	What it is
Callaway & Sant'Anna (2021)	PT-never <i>or</i> PT-not-yet	the plug-in $[\bar{y}_{g,t} - \bar{y}_{g,g-1}] - [\bar{y}_{C,t} - \bar{y}_{C,g-1}]$, $C =$ never- or not-yet-treated
Sun & Abraham (2021)	PT-never	the same never-treated estimand, from a saturated cohort $\times$ event-time regression
Wooldridge (2021); Gardner (2022) <i>did2s</i> ; Borusyak et al. (2024)	PT-all	imputation : fit unit and time effects on untreated cells, average $y - \hat{y}(0)$ over treated cells; identical to each other in balanced panels
Cengiz et al. (2019), stacked	tailored PT	one clean 2×2 per cohort, comparison = units treated $\geq \ell + 1$ periods later
de Chaisemartin & D'Haultfœuille (2020)	PT-not-yet	switchers vs not-yet-switchers, $ATT(g, g)$; treatment may switch off (next chapter)

Sun–Abraham and Callaway–Sant’Anna (never-treated): **one estimand, two computations**. Imputation and Sun–Abraham: **two estimands** — they need not agree.

Which assumption? The real trade-off

PT-all (imputation, extended TWFE)

- uses *all* pre-periods and *all* not-yet-treated units $\Rightarrow$ the most **precise**;
- but imposes parallel *pre*-trends: if they fail, the estimate is **biased**;
- over-identified $\Rightarrow$ pre-trends become a genuine **test**.

PT-never (Sun–Abraham, CS-never)

- comparison group of **fixed composition**; no restriction on pre-trends;
- never uses a unit that chose to adopt *because* of its recent outcomes;
- but never-treated units may be **too different**, or **too few**.

PT-not-yet (CS-not-yet, dCDH)

- more comparison units than PT-never, without restricting pre-trends;
- the **middle ground** the Practitioner's Guide favours in its application;
- composition of the comparison changes over t — report it.

Their advice

“We strongly recommend that researchers clearly state the specific parallel trends assumption they are actually imposing, to allow readers to discuss its plausibility.”

Why estimates differ

Applied to the same data, two heterogeneity-robust estimators generally yield different estimates. Each is the sample counterpart of a different parallel-trends assumption, so a discrepancy points to the assumption on which they disagree:

- Imputation $\neq$ Sun–Abraham: the pre-period is not parallel, so the PT-all assumption is doing work, or the choice of base period matters.
- CS-never $\neq$ CS-not-yet: never-treated and not-yet-treated units trend differently; one of the two is a poor comparison group, and the choice has to be defended.
- Any of them $\neq$ TWFE: effects are dynamic or heterogeneous across cohorts, transmitted into the TWFE coefficient by the forbidden comparisons.

Do not run these estimators as a robustness check

They estimate different parameters under different assumptions: their agreement is not a confirmation, their disagreement not a failure. Choose the assumption you can defend, report the corresponding estimator, and use the others to show which assumption the result depends on.

How the estimators complement each other

A staggered DiD analysis is built in four steps, each with a specific role for the estimators of the previous slides:

- ➊ **The main estimate**, under the parallel-trends assumption chosen for the application — for instance PT-not-yet with base period $g - 1$, estimated by Callaway–Sant’Anna — with the comparison group described.
- ➋ **The evidence for that assumption**: cohort-specific pre-trends against the same comparison group, uniform confidence bands, and a Rambachan–Roth bound giving the largest violation the estimate survives.
- ➌ **The sensitivity analysis**: the same parameter under PT-all, estimated by imputation. Agreement means the additional precision comes at no cost; disagreement indicates a pre-trend problem, to be discussed rather than averaged over.
- ➍ **Heterogeneity** by cohort and by event time, which the aggregate conceals.

The estimators are complements in this precise sense: each one is valid under a different assumption, and comparing their results is a way of assessing those assumptions.

Section 5

Aggregation

From building blocks to one number

$\text{ATT}(g, t)$ are many, noisy, and cohort-specific. Aggregate them with **non-negative, transparent** weights $w_{g,t}$ that sum to one:

$$\text{ATT}^{\text{agg}} = \sum_{g,t} w_{g,t} \text{ATT}(g, t).$$

- **Event-time** aggregation: for each k , average $\text{ATT}(g, g + k)$ over the cohorts observed k periods after treatment, weighted by **cohort size** — the staggered event study.
- **Balanced** event time: keep only cohorts observed at every k in a window, so the composition does not change along the curve.
- **Calendar-time** and **by-cohort** aggregations; an **overall** ATT as the average of post-treatment event times.

Contrast

TWFE is also an aggregation of 2×2 s — with weights chosen by the variance of d_{it} , not by the question, and with forbidden comparisons inside.

Pre-trends in staggered designs

- Under PT-never / PT-not-yet the assumption concerns **post**-treatment periods only: pre-trends measure untreated trend differences in the “wrong” periods — informative, not a test of the assumption. Under **PT-all** they *are* a test.
- **Low power**: joint tests on the leads rarely detect violations of a magnitude that matters (Roth 2022); visual inspection of the leads is not sufficient evidence.
- Report **uniform (sup- t) confidence bands** for the whole event-study curve, not pointwise ones — the did package does by default.
- **Sensitivity**: Rambachan and Roth (2023) bound the effect under violations no larger than the largest pre-period deviation, which turns the inspection of pre-trends into a quantitative statement.
- Cohorts that plausibly **selected on recent outcomes** violate every version: remove them and say so, rather than let them contaminate the comparison.

Section 6

Workflow

What you must build in the data

These estimators need the **timing** structure explicitly:

- the **cohort** g_i — each unit's first treatment period, on *all* of its rows;
- **event time** $k = t - g_i$
- a **never-treated** indicator (or the last-treated cohort as never-treated, dropping the periods where everyone is treated);
- units treated in the **first** period have no pre-period: drop them.

Software builds the 2×2 s and the aggregation for you — you supply g_i and the **assumption**.

In Python and R

```
import pyfixest as pf
# PT-all: imputation / two-stage DiD (Gardner 2022)
m = pf.did2s(df, yname="y", first_stage=~ 0 | id + t",
             second_stage=~ i(k, ref=-1)", treatment="d",
             cluster="id")
# PT-never: saturated cohort x event-time regression (Sun-Abraham)
es = pf.event_study(df, yname="y", idname="id", tname="t",
                    gname="g", estimator="saturated")

# PT-not-yet: Callaway-Sant'Anna in R (did), base period g-1
att <- did::att_gt(yname="y", tname="t", idname="id", gname="g",
                  data=df, control_group="notyettreated")
did::aggte(att, type="dynamic")      # event-time aggregation
```

- Each line embodies an assumption; write it next to the estimate you report.

The forward-engineering workflow (Baker et al. 2026)

- ➊ **Define the target** — $ATT(g, t)$ and the aggregation you care about.
- ➋ **State the identifying assumptions** — which parallel trends (never / not-yet / all), no-anticipation; argue them and bring pre-period evidence.
- ➌ **Choose the estimator** that is the sample analogue of those assumptions.
- ➍ **Discuss the source of uncertainty** — sampling or design-based; cluster accordingly.
- ➎ **Estimate.**
- ➏ **Sensitivity analysis** — bounds on parallel-trends violations; the same target under the neighbouring assumption.
- ➐ **Heterogeneity** — by cohort, by event time, by group.
- ➑ **Keep learning** — if the assumptions are not credible, change design, not estimator.

On TWFE

“It has well-understood, potentially serious, and easily remedied problems, and we do not recommend using it.” — Baker et al. (2026)

Takeaways

- With staggered adoption, TWFE is a **weighted average of 2×2 DiDs** including **forbidden** ones (already-treated as control); dynamics alone bias it, up to a **sign reversal**. The dynamic version contaminates the leads.
- The object is $ATT(g, t)$; identification is a **choice of comparison group and base period**, i.e. of **one of three parallel-trends assumptions**.
- Callaway–Sant’Anna, Sun–Abraham, imputation, stacked, dCDH are **sample analogues of those assumptions** — not five robustness checks. Disagreement is evidence about the assumptions.
- Aggregate with transparent non-negative weights; report cohort-level pre-trends, uniform bands, a sensitivity bound, and **the assumption you impose**.

Next

Beyond the switch — treatments that are a *dose*, or that turn *on and off*.

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